

Creation Date 23-Sep-2009

Revision Date 06-Dec-2024

Revision Number 11

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: Zinc chloride, 0.7M solution in THF  
Cat No. : 389450000; 389451000; 389458000

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

## Physical hazards

Flammable liquids

Category 2 (H225)

## Health hazards

Acute oral toxicity

Category 4 (H302)

Skin Corrosion/Irritation

Category 1 B (H314)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Carcinogenicity

Category 2 (H351)

Specific target organ toxicity - (single exposure)

Category 3 (H335) (H336)

## Environmental hazards

Acute aquatic toxicity

Category 1 (H400)

Chronic aquatic toxicity

Category 1 (H410)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

## Hazard Statements

H225 - Highly flammable liquid and vapor

H302 - Harmful if swallowed

H314 - Causes severe skin burns and eye damage

H335 - May cause respiratory irritation

H336 - May cause drowsiness or dizziness

H351 - Suspected of causing cancer

H410 - Very toxic to aquatic life with long lasting effects

EUH019 - May form explosive peroxides

## Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

## 2.3. Other hazards

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

## Section 3: Composition/information on ingredients

### 3.2. Mixtures

| Component       | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                          |
|-----------------|-----------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | 109-99-9  | 203-726-8         | 90       | Flam. Liq. 2 (H225)<br>Acute Tox. 4 (H302)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)<br>STOT SE 3 (H336)<br>Carc. 2 (H351)<br>(EUH019)    |
| Zinc chloride   | 7646-85-7 | EEC No. 231-592-0 | 10       | Acute Tox. 4 (H302)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>STOT SE 3 (H335)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |

| Component       | Specific concentration limits (SCL's)                                    | M-Factor                  | Component notes |
|-----------------|--------------------------------------------------------------------------|---------------------------|-----------------|
| Tetrahydrofuran | Acute Tox. 4 :: C>82.5%<br>Eye Irrit. 2 :: C>=25%<br>STOT SE 3 :: C>=25% | -                         | -               |
| Zinc chloride   | STOT SE 3 (H335) :: C>=5%                                                | Acute = 10<br>Chronic = 1 | -               |

| Components      | Reach Registration Number |
|-----------------|---------------------------|
| Tetrahydrofuran | 01-2119444314-46          |
| Zinc chloride   | 01-2119472431-44          |

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                        |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                           |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

## 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Causes central nervous system depression

## 4.3. Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## Section 5: Firefighting measures

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Fumes, Chlorine, Zinc.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep away from heat, sparks and flame. Flammables area. Shelf life 12 months. May form explosive peroxides on prolonged storage. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Store under an inert atmosphere. To maintain product quality: Keep refrigerated. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component       | European Union                                                                                                              | The United Kingdom                                                                                                        | France                                                                                                                                                                                                                 | Belgium                                                                                                                               | Spain                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm (8h)<br>TWA: 150 mg/m <sup>3</sup> (8h)<br>STEL: 100 ppm (15min)<br>STEL: 300 mg/m <sup>3</sup> (15min)<br>Skin | STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>Skin | TWA / VME: 50 ppm (8 heures). restrictive limit<br>TWA / VME: 150 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 100 ppm. restrictive limit<br>STEL / VLCT: 300 mg/m <sup>3</sup> . restrictive limit | TWA: 50 ppm 8 uren<br>TWA: 150 mg/m <sup>3</sup> 8 uren<br>STEL: 100 ppm 15 minuten<br>STEL: 300 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 100 ppm (15 minutos).<br>STEL / VLA-EC: 300 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 50 ppm (8 horas)<br>TWA / VLA-ED: 150 mg/m <sup>3</sup> (8 horas)<br>Piel |

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

|               |  |                                                                   |                                            |                                                                         |                                                                                                 |
|---------------|--|-------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|               |  |                                                                   | Peau                                       |                                                                         |                                                                                                 |
| Zinc chloride |  | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr | TWA / VME: 1 mg/m <sup>3</sup> (8 heures). | TWA: 1 mg/m <sup>3</sup> 8 uren<br>STEL: 2 mg/m <sup>3</sup> 15 minuten | STEL / VLA-EC: 2 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 1 mg/m <sup>3</sup> (8 horas) |

| Component       | Italy                                                                                                                                                                                                          | Germany                                                                                                                                                                                                                                                            | Portugal                                                                                                                                | The Netherlands                                                                                                                        | Finland                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm 8 ore.<br>Time Weighted Average<br>TWA: 150 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>STEL: 100 ppm 15 minuti. Short-term<br>STEL: 300 mg/m <sup>3</sup> 15 minuti. Short-term<br>Pelle | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 150 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 20 ppm (8 Stunden). MAK<br>TWA: 60 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 40 ppm<br>Höhepunkt: 120 mg/m <sup>3</sup><br>Haut | STEL: 100 ppm 15 minutos<br>STEL: 300 mg/m <sup>3</sup> 15 minutos<br>TWA: 50 ppm 8 horas<br>TWA: 150 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>STEL: 200 ppm 15 minuten<br>STEL: 600 mg/m <sup>3</sup> 15 minuten<br>TWA: 100 ppm 8 uren<br>TWA: 300 mg/m <sup>3</sup> 8 uren | TWA: 50 ppm 8 tunteina<br>TWA: 150 mg/m <sup>3</sup> 8 tunteina<br>STEL: 100 ppm 15 minuutteina<br>STEL: 300 mg/m <sup>3</sup> 15 minuutteina<br>Iho |
| Zinc chloride   |                                                                                                                                                                                                                | TWA: 0.1 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 2 mg/m <sup>3</sup><br>Höhepunkt: 0.4 mg/m <sup>3</sup><br>Höhepunkt: 4 mg/m <sup>3</sup>                                                                   | STEL: 2 mg/m <sup>3</sup> 15 minutos<br>TWA: 1 mg/m <sup>3</sup> 8 horas                                                                |                                                                                                                                        | TWA: 1 mg/m <sup>3</sup> 8 tunteina                                                                                                                  |

| Component       | Austria                                                                                                                                                     | Denmark                                                                                                                                  | Switzerland                                                                                                                                      | Poland                                                                            | Norway                                                                                                                                                                        |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | Haut<br>MAK-KZGW: 100 ppm 15 Minuten<br>MAK-KZGW: 300 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 50 ppm 8 Stunden<br>MAK-TMW: 150 mg/m <sup>3</sup> 8 Stunden | TWA: 50 ppm 8 timer<br>TWA: 150 mg/m <sup>3</sup> 8 timer<br>STEL: 300 mg/m <sup>3</sup> 15 minutter<br>STEL: 100 ppm 15 minutter<br>Hud | Haut/Peau<br>STEL: 100 ppm 15 Minuten<br>STEL: 300 mg/m <sup>3</sup> 15 Minuten<br>TWA: 50 ppm 8 Stunden<br>TWA: 150 mg/m <sup>3</sup> 8 Stunden | STEL: 300 mg/m <sup>3</sup> 15 minutach<br>TWA: 150 mg/m <sup>3</sup> 8 godzinach | TWA: 50 ppm 8 timer<br>TWA: 150 mg/m <sup>3</sup> 8 timer<br>STEL: 75 ppm 15 minutter. value calculated<br>STEL: 187.5 mg/m <sup>3</sup> 15 minutter. value calculated<br>Hud |
| Zinc chloride   |                                                                                                                                                             | TWA: 0.5 mg/m <sup>3</sup> 8 timer<br>STEL: 1 mg/m <sup>3</sup> 15 minutter                                                              | TWA: 1 mg/m <sup>3</sup> 8 Stunden                                                                                                               | STEL: 2 mg/m <sup>3</sup> 15 minutach<br>TWA: 1 mg/m <sup>3</sup> 8 godzinach     | TWA: 1 mg/m <sup>3</sup> 8 timer<br>STEL: 3 mg/m <sup>3</sup> 15 minutter. value calculated                                                                                   |

| Component       | Bulgaria                                                                                                           | Croatia                                                                                                                                                           | Ireland                                                                                                                     | Cyprus                                                                                                                               | Czech Republic                                                                                                 |
|-----------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50.0 ppm<br>TWA: 150.0 mg/m <sup>3</sup><br>STEL : 100 ppm<br>STEL : 300.0 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 50 ppm 8 satima.<br>TWA-GVI: 150 mg/m <sup>3</sup> 8 satima.<br>STEL-KGVI: 100 ppm 15 minutama.<br>STEL-KGVI: 300 mg/m <sup>3</sup> 15 minutama. | TWA: 50 ppm 8 hr.<br>TWA: 150 mg/m <sup>3</sup> 8 hr.<br>STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min<br>Skin | Skin-potential for cutaneous absorption<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 150 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 300 mg/m <sup>3</sup> |
| Zinc chloride   |                                                                                                                    | TWA-GVI: 1 mg/m <sup>3</sup> 8 satima. fume<br>STEL-KGVI: 2 mg/m <sup>3</sup> 15 minutama. fume                                                                   | TWA: 1 mg/m <sup>3</sup> 8 hr. fume<br>STEL: 2 mg/m <sup>3</sup> 15 min                                                     |                                                                                                                                      | TWA: 1 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 2 mg/m <sup>3</sup>                                           |

| Component       | Estonia                                                                                                                                               | Gibraltar                                                                                                                          | Greece                                                                                     | Hungary                                                                                                                                                                   | Iceland                                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | Nahk<br>TWA: 50 ppm 8 tundides.<br>TWA: 150 mg/m <sup>3</sup> 8 tundides.<br>STEL: 100 ppm 15 minutites.<br>STEL: 300 mg/m <sup>3</sup> 15 minutites. | Skin notation<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min | STEL: 250 ppm<br>STEL: 735 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup> | STEL: 300 mg/m <sup>3</sup> 15 percekben. CK<br>STEL: 100 ppm 15 percekben. CK<br>TWA: 150 mg/m <sup>3</sup> 8 órában. AK<br>TWA: 50 ppm 8 órában. AK<br>lehetséges borón | STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm 8 klukkustundum.<br>TWA: 150 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation |

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

|               |                                                            |  |                                                       |                                                                       |                                                                                                                          |
|---------------|------------------------------------------------------------|--|-------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Zinc chloride | TWA: 1 mg/m <sup>3</sup> 8<br>tundides. respirable<br>dust |  | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | keresztüli felszívódás<br>TWA: 1 mg/m <sup>3</sup> 8<br>órában. AK Zn | TWA: 0.5 mg/m <sup>3</sup> 8<br>klukkustundum. Zn<br>including fume<br>Ceiling: 1 mg/m <sup>3</sup> Zn<br>including fume |
|---------------|------------------------------------------------------------|--|-------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|

| Component       | Latvia                                                                                                                                  | Lithuania                                                                                                  | Luxembourg                                                                                                                                                                                               | Malta                                                                                                                                                                        | Romania                                                                                                                                          |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | skin - potential for<br>cutaneous exposure<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 50 ppm IPRD<br>TWA: 150 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup> | Possibility of significant<br>uptake through the skin<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 150 mg/m <sup>3</sup> 8<br>Stunden<br>STEL: 100 ppm 15<br>Minuten<br>STEL: 300 mg/m <sup>3</sup> 15<br>Minuten | possibility of significant<br>uptake through the skin<br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup><br>STEL: 100 ppm 15<br>minuti<br>STEL: 300 mg/m <sup>3</sup> 15<br>minuti | Skin notation<br>TWA: 50 ppm 8 ore<br>TWA: 150 mg/m <sup>3</sup> 8 ore<br>STEL: 100 ppm 15<br>minute<br>STEL: 300 mg/m <sup>3</sup> 15<br>minute |
| Zinc chloride   |                                                                                                                                         | TWA: 1 mg/m <sup>3</sup><br>respirable fraction IPRD<br>Oda                                                |                                                                                                                                                                                                          |                                                                                                                                                                              |                                                                                                                                                  |

| Component       | Russia                     | Slovak Republic                                                                                                      | Slovenia                                                                                                                                    | Sweden                                                                                                                                                                  | Turkey                                                                                                                                    |
|-----------------|----------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | MAC: 100 mg/m <sup>3</sup> | Ceiling: 300 mg/m <sup>3</sup><br>Potential for cutaneous<br>absorption<br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 50 ppm 8 urah<br>TWA: 150 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 100 ppm 15<br>minutah<br>STEL: 300 mg/m <sup>3</sup> 15<br>minutah | Binding STEL: 100 ppm<br>15 minuter<br>Binding STEL: 300<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 50 ppm 8 timmar.<br>NGV<br>TLV: 150 mg/m <sup>3</sup> 8<br>timmar. NGV | Deri<br>TWA: 50 ppm 8 saat<br>TWA: 150 mg/m <sup>3</sup> 8 saat<br>STEL: 100 ppm 15<br>dakika<br>STEL: 300 mg/m <sup>3</sup> 15<br>dakika |
| Zinc chloride   |                            |                                                                                                                      |                                                                                                                                             | TLV: 1 mg/m <sup>3</sup> 8 timmar.<br>NGV                                                                                                                               |                                                                                                                                           |

## Biological limit values

List source(s):

| Component       | European Union | United Kingdom | France | Spain                                         | Germany                                          |
|-----------------|----------------|----------------|--------|-----------------------------------------------|--------------------------------------------------|
| Tetrahydrofuran |                |                |        | Tetrahydrofuran: 2 mg/L<br>urine end of shift | Tetrahydrofuran: 2 mg/L<br>urine (end of shift ) |

| Component       | Gibraltar | Latvia | Slovak Republic                                                   | Luxembourg | Turkey |
|-----------------|-----------|--------|-------------------------------------------------------------------|------------|--------|
| Tetrahydrofuran |           |        | Tetrahydrofuran: 2 mg/L<br>urine end of exposure or<br>work shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

MDHS 99 Metals in air by ICP-AES

MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component     | Acute effects local<br>(Oral) | Acute effects<br>systemic (Oral) | Chronic effects local<br>(Oral) | Chronic effects<br>systemic (Oral) |
|---------------|-------------------------------|----------------------------------|---------------------------------|------------------------------------|
| Zinc chloride |                               |                                  |                                 | 0.83 mg/kg bw/day                  |

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

|                  |  |  |  |  |
|------------------|--|--|--|--|
| 7646-85-7 ( 10 ) |  |  |  |  |
|------------------|--|--|--|--|

| Component                          | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) |                              |                                 |                                | DNEL = 12.6mg/kg<br>bw/day        |
| Zinc chloride<br>7646-85-7 ( 10 )  |                              |                                 |                                | DNEL = 8.3mg/kg<br>bw/day         |

| Component                          | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | DNEL = 300mg/m <sup>3</sup>      | DNEL = 96mg/m <sup>3</sup>          | DNEL = 150mg/m <sup>3</sup>        | DNEL = 72.4mg/m <sup>3</sup>          |
| Zinc chloride<br>7646-85-7 ( 10 )  |                                  |                                     |                                    | DNEL = 1mg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                          | Fresh water     | Fresh water sediment             | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)          |
|------------------------------------|-----------------|----------------------------------|--------------------|------------------------------------|-----------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | PNEC = 4.32mg/L | PNEC = 23.3mg/kg<br>sediment dw  | PNEC = 21.6mg/L    | PNEC = 4.6mg/L                     | PNEC = 2.13mg/kg<br>soil dw |
| Zinc chloride<br>7646-85-7 ( 10 )  | PNEC = 20.6µg/L | PNEC = 117.8mg/kg<br>sediment dw |                    | PNEC = 100µg/L                     | PNEC = 35.6mg/kg<br>soil dw |

| Component                          | Marine water     | Marine water sediment           | Marine water Intermittent | Food chain             | Air |
|------------------------------------|------------------|---------------------------------|---------------------------|------------------------|-----|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | PNEC = 0.432mg/L | PNEC = 2.33mg/kg<br>sediment dw |                           | PNEC = 67mg/kg<br>food |     |
| Zinc chloride<br>7646-85-7 ( 10 )  | PNEC = 6.1µg/L   | PNEC = 56.5mg/kg<br>sediment dw |                           |                        |     |

## 8.2. Exposure controls

### Engineering Measures

Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material  | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Butyl rubber    | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Neoprene gloves |                                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.



# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Organic gases and vapours filter Type A Brown conforming to EN14387                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Odor</b>                                    | Irritating               |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 66 °C / 150.8 °F         |                                          |
| <b>Flammability (liquid)</b>                   | Highly flammable         | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | -22 °C / -7.6 °F         | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Miscible                 |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| <b>Tetrahydrofuran</b>                         | 0.45                     |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 0.980                    |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No information available | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | (liquid) Not applicable  |                                          |

### 9.2. Other information

|                             |                                             |
|-----------------------------|---------------------------------------------|
| <b>Explosive Properties</b> | Vapors may form explosive mixtures with air |
|-----------------------------|---------------------------------------------|

## Section 10: Stability and reactivity

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

## 10.1. Reactivity

Yes

## 10.2. Chemical stability

May form explosive peroxides. Stable under normal conditions.

## 10.3. Possibility of hazardous reactions

**Hazardous Polymerization**  
**Hazardous Reactions**

Hazardous polymerization may occur.  
None under normal processing.

## 10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.

## 10.5. Incompatible materials

Bases. Strong oxidizing agents. Strong acids.

## 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Fumes. Chlorine. Zinc.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

#### Toxicology data for the components

| Component       | LD50 Oral          | LD50 Dermal           | LC50 Inhalation                               |
|-----------------|--------------------|-----------------------|-----------------------------------------------|
| Tetrahydrofuran | 1650 mg/kg ( Rat ) | > 2000 mg/kg (Rabbit) | 180 mg/L ( Rat ) 1 h<br>53.9 mg/L ( Rat ) 4 h |
| Zinc chloride   | 350 mg/kg (Rat)    | -                     | LC50 <= 1975 mg/m <sup>3</sup> ( Rat ) 10 min |

##### (b) skin corrosion/irritation;

Category 1 B

##### (c) serious eye damage/irritation;

Category 1

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                          | Test method                                       | Test species | Study result    |
|------------------------------------|---------------------------------------------------|--------------|-----------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | Local Lymph Node Assay<br>OECD Test Guideline 429 | mouse        | non-sensitising |

##### (e) germ cell mutagenicity;

No data available

| Component       | Test method             | Test species | Study result |
|-----------------|-------------------------|--------------|--------------|
| Tetrahydrofuran | OECD Test Guideline 476 | in vivo      | negative     |

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

|                 |                                                                                        |                                             |                   |
|-----------------|----------------------------------------------------------------------------------------|---------------------------------------------|-------------------|
| 109-99-9 ( 90 ) | Gene cell mutation<br>-----<br>OECD Test Guideline 473<br>Chromosomal aberration assay | Mammalian<br>-----<br>in vitro<br>Mammalian | -----<br>negative |
|-----------------|----------------------------------------------------------------------------------------|---------------------------------------------|-------------------|

(f) carcinogenicity;

Category 2

Limited evidence of a carcinogenic effect

| Component       | EU | UK | Germany | IARC     |
|-----------------|----|----|---------|----------|
| Tetrahydrofuran |    |    |         | Group 2B |

(g) reproductive toxicity;

No data available

| Component                          | Test method             | Test species / Duration | Study result      |
|------------------------------------|-------------------------|-------------------------|-------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | OECD Test Guideline 416 | Rat<br>2 Generation     | NOAEL = 3,000 ppm |

(h) STOT-single exposure;

Category 3

Results / Target organs

Respiratory system, Central nervous system (CNS).

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

No data available

Other Adverse Effects

The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Causes central nervous system depression.

## 11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component       | Freshwater Fish                                                                     | Water Flea                                   | Freshwater Algae           |
|-----------------|-------------------------------------------------------------------------------------|----------------------------------------------|----------------------------|
| Tetrahydrofuran | 2160 mg/l LC50 = 96 h<br>Pimephales promelas<br>Leuciscus idus: LC50: 2820 mg/L/48h | EC50 48 h 3485 mg/l<br>EC50: >10000 mg/L/24h |                            |
| Zinc chloride   | LC50: 0.4-2.2 mg/L/96h<br>(Cyprinus carpio)                                         | EC50: 0.2 mg/L/48h                           | EC50: 0.027-0.105 mg/L/72h |

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

| Component     | Microtox | M-Factor                  |
|---------------|----------|---------------------------|
| Zinc chloride |          | Acute = 10<br>Chronic = 1 |

**12.2. Persistence and degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary based on information available, May persist.  
**Persistence**  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**12.3. Bioaccumulative potential** May have some potential to bioaccumulate

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Tetrahydrofuran | 0.45    | No data available             |
| Zinc chloride   |         | 16000 dimensionless           |

**12.4. Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

**12.6. Endocrine disrupting properties**  
**Endocrine Disruptor Information**

| Component       | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|-----------------|------------------------------------------|--------------------------------------------------|
| Tetrahydrofuran | Group III Chemical                       |                                                  |

**12.7. Other adverse effects**  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

**Switzerland - Waste Ordinance** Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance,

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## Section 14: Transport information

### IMDG/IMO

**14.1. UN number** UN2924  
**14.2. UN proper shipping name** Flammable liquid, corrosive, n.o.s.  
**Technical Shipping Name** Tetrahydrofuran, Zinc chloride  
**14.3. Transport hazard class(es)** 3  
**Subsidiary Hazard Class** 8  
**14.4. Packing group** II

### ADR

**14.1. UN number** UN2924  
**14.2. UN proper shipping name** Flammable liquid, corrosive, n.o.s.  
**Technical Shipping Name** Tetrahydrofuran, Zinc chloride  
**14.3. Transport hazard class(es)** 3  
**Subsidiary Hazard Class** 8  
**14.4. Packing group** II

### IATA

**14.1. UN number** UN2924  
**14.2. UN proper shipping name** Flammable liquid, corrosive, n.o.s.  
**Technical Shipping Name** Tetrahydrofuran, Zinc chloride  
**14.3. Transport hazard class(es)** 3  
**Subsidiary Hazard Class** 8  
**14.4. Packing group** II

**14.5. Environmental hazards** Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Tetrahydrofuran | 109-99-9  | 203-726-8 | -      | -   | X     | X    | KE-33454 | X    | X    |
| Zinc chloride   | 7646-85-7 | 231-592-0 | -      | -   | X     | X    | KE-35535 | X    | X    |

| Component       | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Tetrahydrofuran | 109-99-9  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Zinc chloride   | 7646-85-7 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component       | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | 109-99-9  | -                                                                   | Use restricted. See entry 75.<br>(see link for restriction details)           | -                                                                                                     |
| Zinc chloride   | 7646-85-7 | -                                                                   | Use restricted. See entry 75.<br>(see link for restriction details)           | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component       | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Tetrahydrofuran | 109-99-9  | Not applicable                                                                            | Not applicable                                                                           |
| Zinc chloride   | 7646-85-7 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 3 (self classification)

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------|---------------------------------------|-------------------------|
| Tetrahydrofuran | WGK1                                  |                         |
| Zinc chloride   | WGK3                                  |                         |

| Component       | France - INRS (Tables of occupational diseases)      |
|-----------------|------------------------------------------------------|
| Tetrahydrofuran | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

| Component                          | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects  
EUH019 - May form explosive peroxides  
H225 - Highly flammable liquid and vapor  
H319 - Causes serious eye irritation

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

# SAFETY DATA SHEET

Zinc chloride, 0.7M solution in THF

Revision Date 06-Dec-2024

---

Chemical incident response training.

**Creation Date** 23-Sep-2009  
**Revision Date** 06-Dec-2024  
**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**